Haiti Earthquake Data Analysis

Patterns in Crisis Reporting and Aid Needs

서울과학기술대학교
SEOULTECH

Jostin
Department of Data Science

2025.05.30

About the Dataset

Haiti Dataset

- The dataset is about the reported incidents in the **Ushahidi** platform by the Inter-Disciplinary Unit for Disaster Risk Reduction (IDU) during or after the **2010 Haiti earthquake**.
- **Ushahidi**
  - Ushahidi is an open-source crisis mapping platform originally developed in Kenya in 2008 to map reports of violence via crowdsourced information (SMS, email, social media, etc.).
  - It allows users to report incidents, which are then mapped in real time.
- **2010 Haiti Earthquake**
  - A devastating earthquake struck Haiti on January 12, 2010.
  - The global response included the use of digital humanitarian tools, especially for crowdsourcing real-time information.
- The deployment of Ushahidi in Haiti helped collect, verify, and visualize:
  - SOS messages from victims.
  - Requests for aid.
  - Damage reports.
- No. of records: 3593

Reason for Choosing

Haiti Dataset – Earthquake Disaster response dataset

Opportunity to apply and explore what we learned in class:

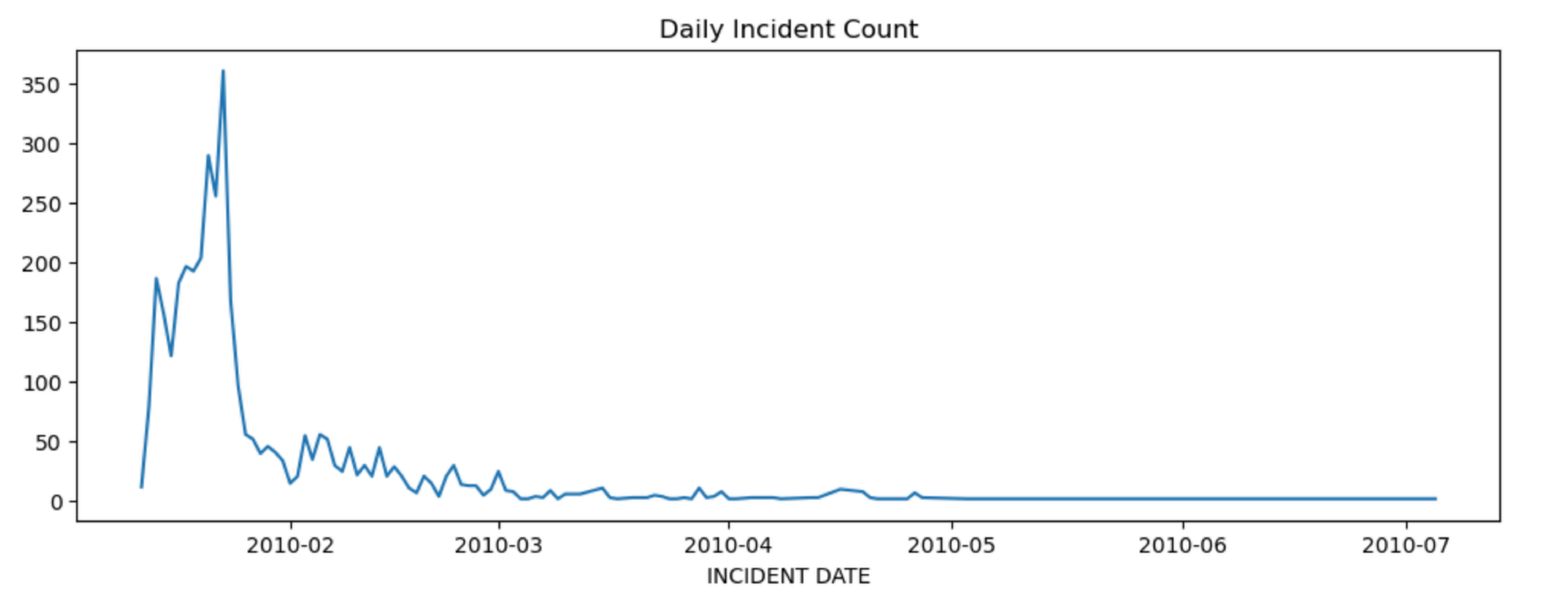
- Has a distinct and unique purpose – very good for **Visual Data Storytelling.**
- The dataset offers a various types of data to explore for Data Analysis.
    - Data itself can stand on its own and can be interpreted individually.
    - Some of data can be refined to produce more insights
    - Variable Data are closely related to each other yet aggregated with other variables, provide a new interpretation of the data.

Objective of the Analysis

- The objective of the analysis is to answer the following questions:
  - What kinds of Incidents were reported during the 2012 Haiti Earthquake Disaster?
    - What trends can be observed in the frequency and category of incidents over time?
  - When strong and devastative earthquakes occur what kind of aid/help should we expect to be requested and should organizations give out as aid to the victims?
    - Within a month after the earthquake?
    - After 3 months after the earthquake?

Descriptive Analysis

Incidents Reported



The Daily Incident Count was highest during the first month after the earthquake occured and have fell significantly in the following months.

Descriptive Analysis

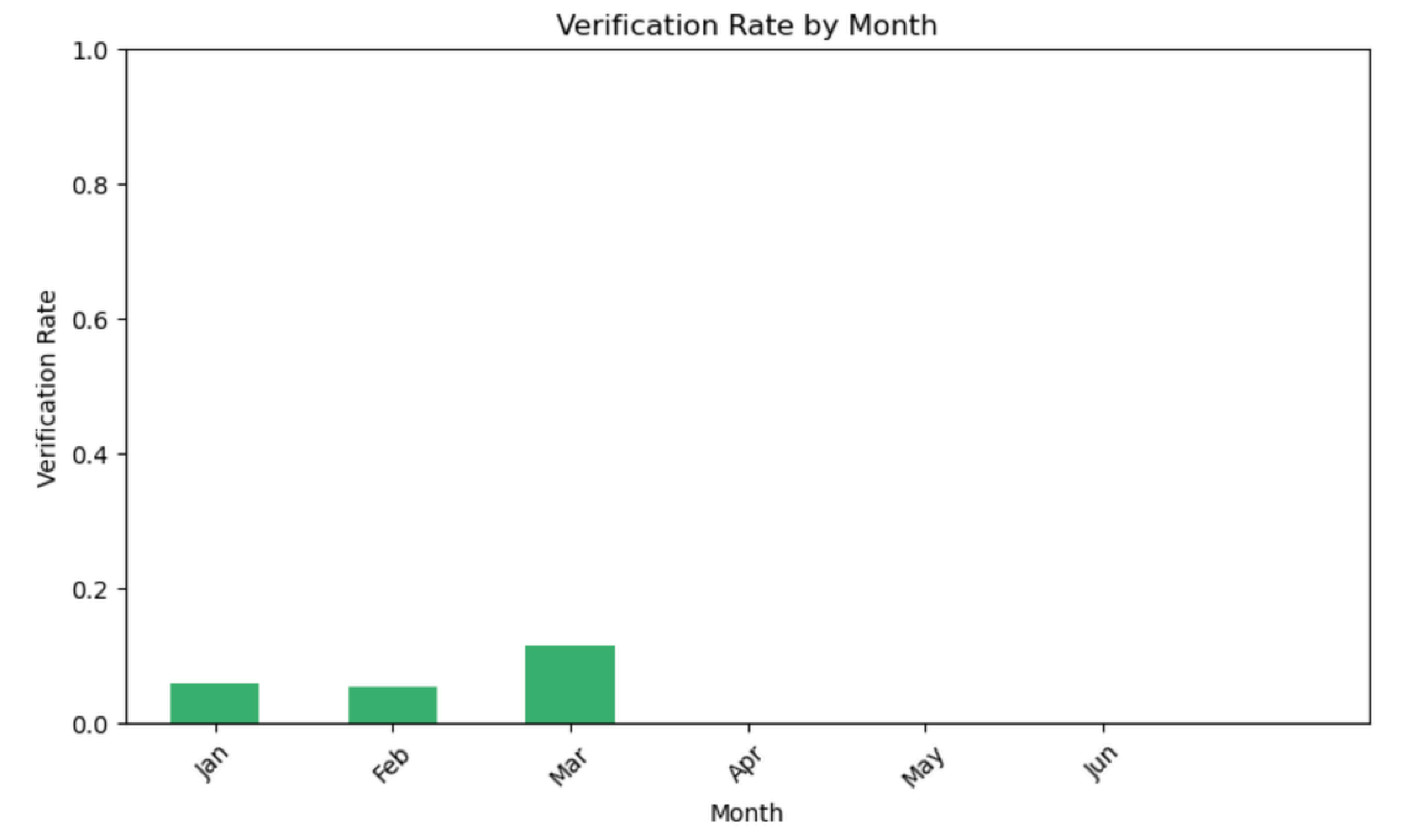
Incidents Reported by Category

Category	Frequency
2a. Penurie d'aliments | Food Shortage	1597
2b. Penurie d'eau | Water shortage	1333
2. Urgences logistiques | Vital Lines	498
2d. Refuge | Shelter needed	477
1. Urgences | Emergency	350
7a. Distribution d'aliments | Food distributio...	333
3c. Besoins en materiels et medicaments | Medi...	305
7. Secours | Services Available	304
8e. Nouvelles de Personnes | Persons News	295
7d. Services de sante | Hospital/Clinics Opera...	258

The Food Shortage category recorded the highest number Incidents / Requests, followed by Water Shortage and Vital lines in the Top 3. This means that during an earthquake disaster (especially a strong one), basic necessities like food, and communication will be top priority needs.

Descriptive Analysis

Verification of Reported Incidents

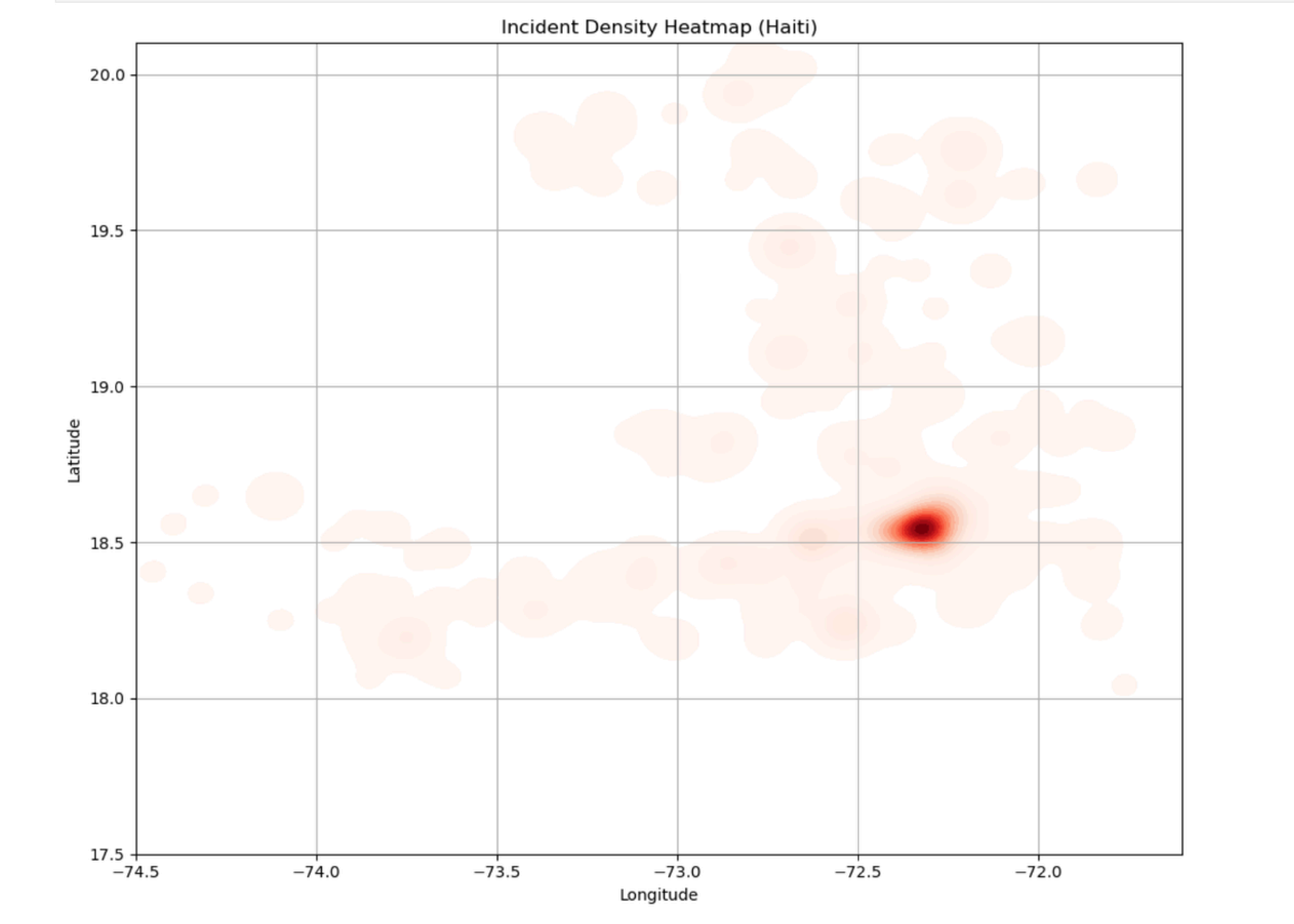


VERIFIED	NO	YES
CATEGORY_clean		
1. Urgences | Emergency	94.3	5.7
1a. Highly vulnerable	100.0	0.0
1b. Urgence medicale | Medical Emergency	93.8	6.2
1c. Personnes prises au piege | People trapped	97.6	2.4
1d. Incendie | Fire	85.7	14.3
2. Urgences logistiques | Vital Lines	94.9	5.1
2a. Penurie d'aliments | Food Shortage	94.0	6.0
2b. Penurie d'eau | Water shortage	93.2	6.8
2c. Eau contaminee | Contaminated water	77.8	22.2
2c. Probleme de securite | Security Concern	100.0	0.0

The Verification rate is generally not high and has only lasted until March. Indicating that there were no verification done starting April.

Almost all of the incidents are not verified, an indication that raises quality control on the incident reporting system at that time.

Descriptive Analysis



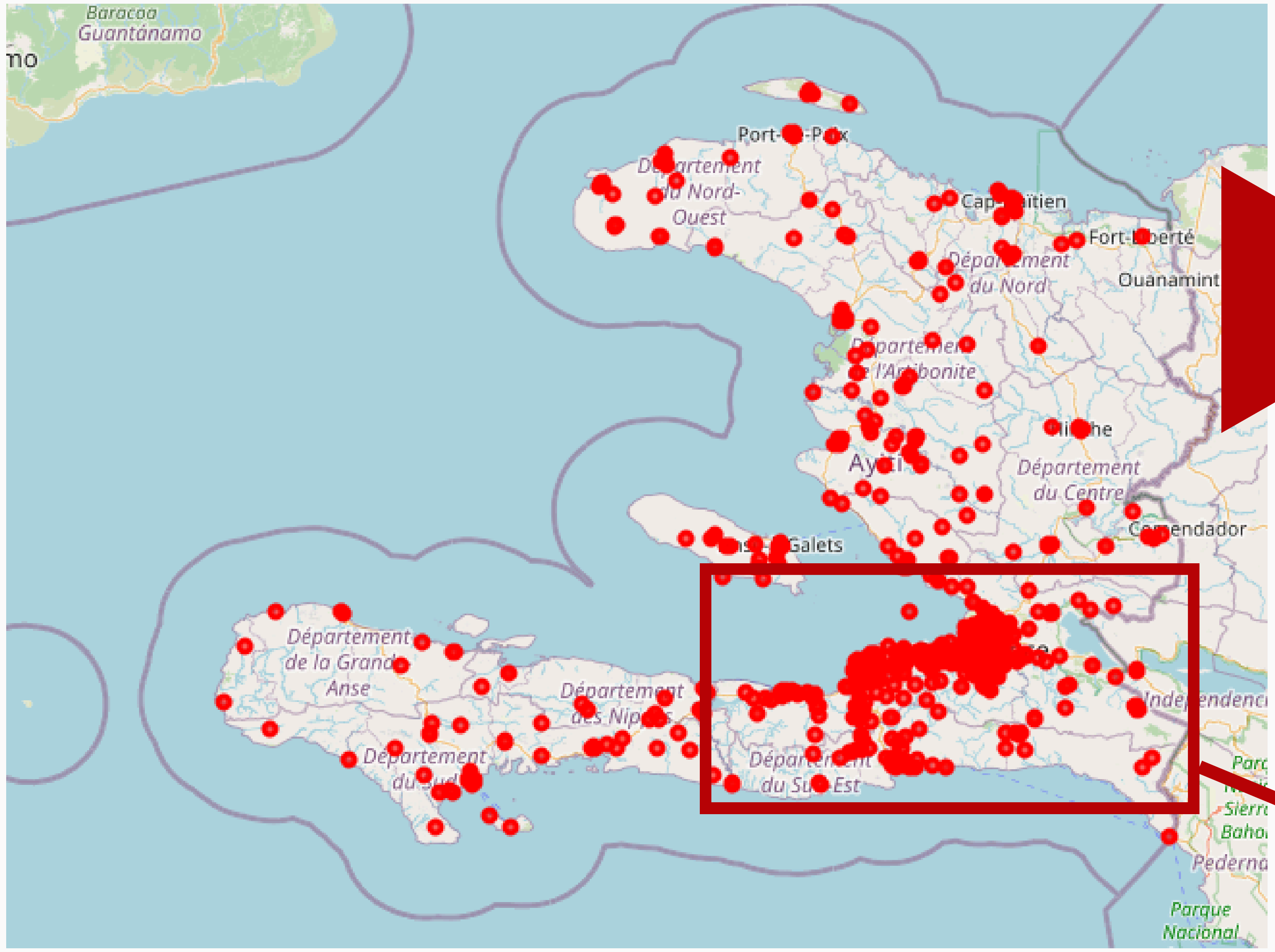
*for this analysis, invalid data (coordinates outside Haiti) were removed improve accuracy of the graph

Incidents per Location

Although there were recorded incidents allover Haiti, it seems most of the incidents are concentrated in one area.

Probably this is where the epicenter of the earthquake was.

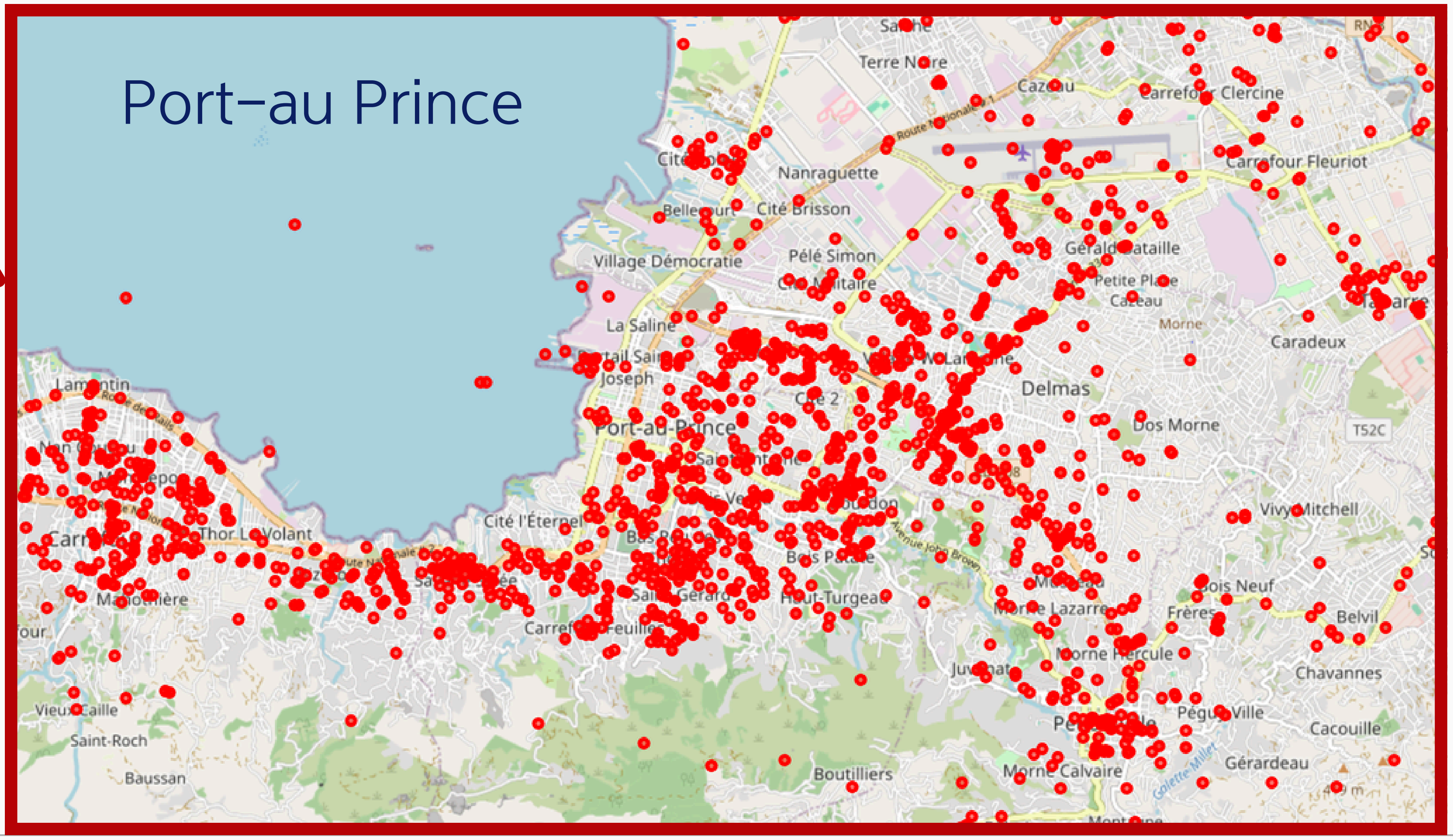
Descriptive Analysis



In the Haiti Plot Map, we can see that there incidents reported outside of Port-au Prince. Indicating that the earthquake has affected other regions as well.

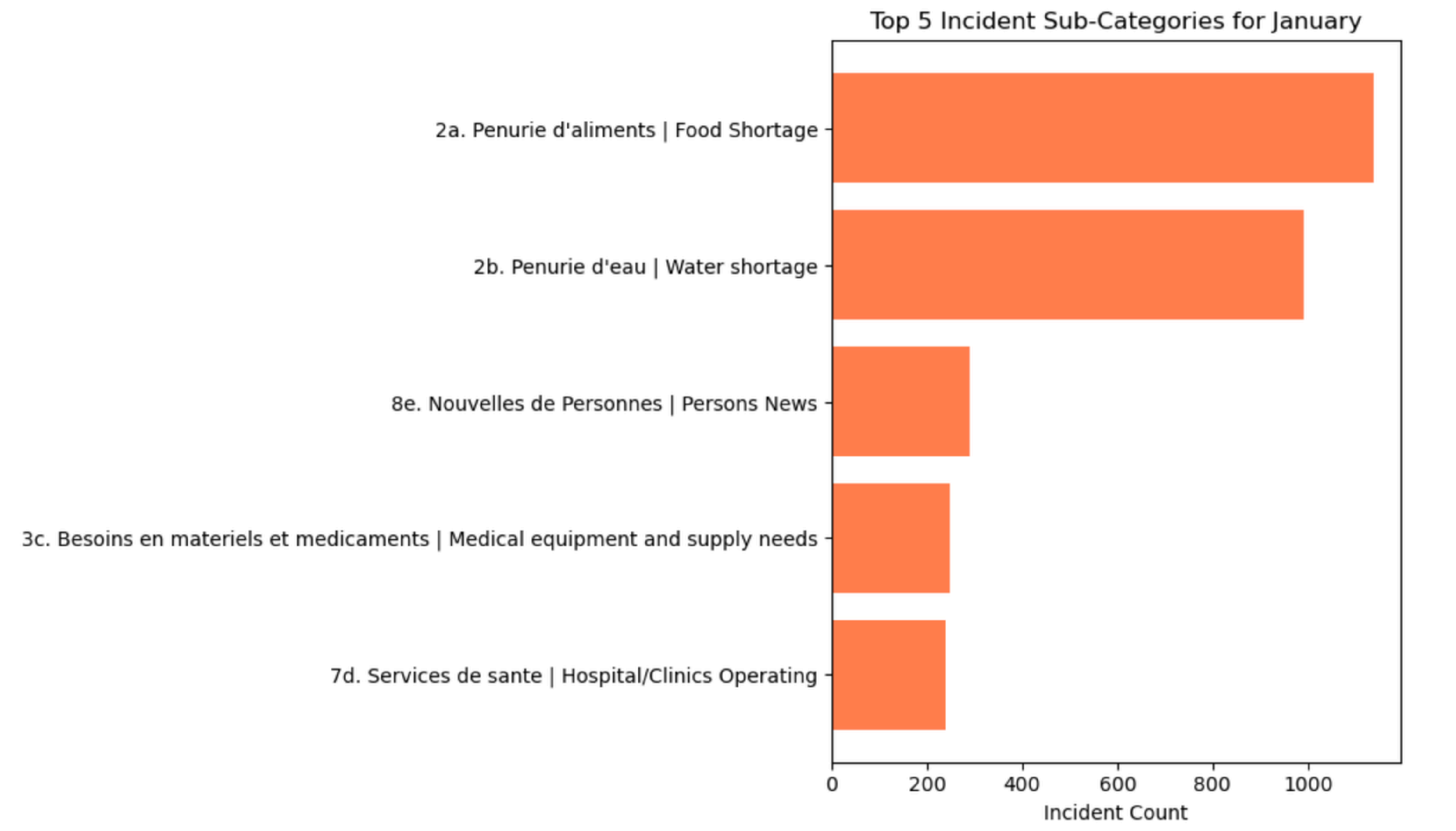
Incidents plotted on Haiti Map

Incidents plotted per location using the latitude and longitude data. Incidents were plotted across Haiti. However, majority of the incidents were located near Port-au Prince, the earthquake's epicenter.



Prescriptive Analysis

Top 5 Incidents for the Month of January (Sub–Category)

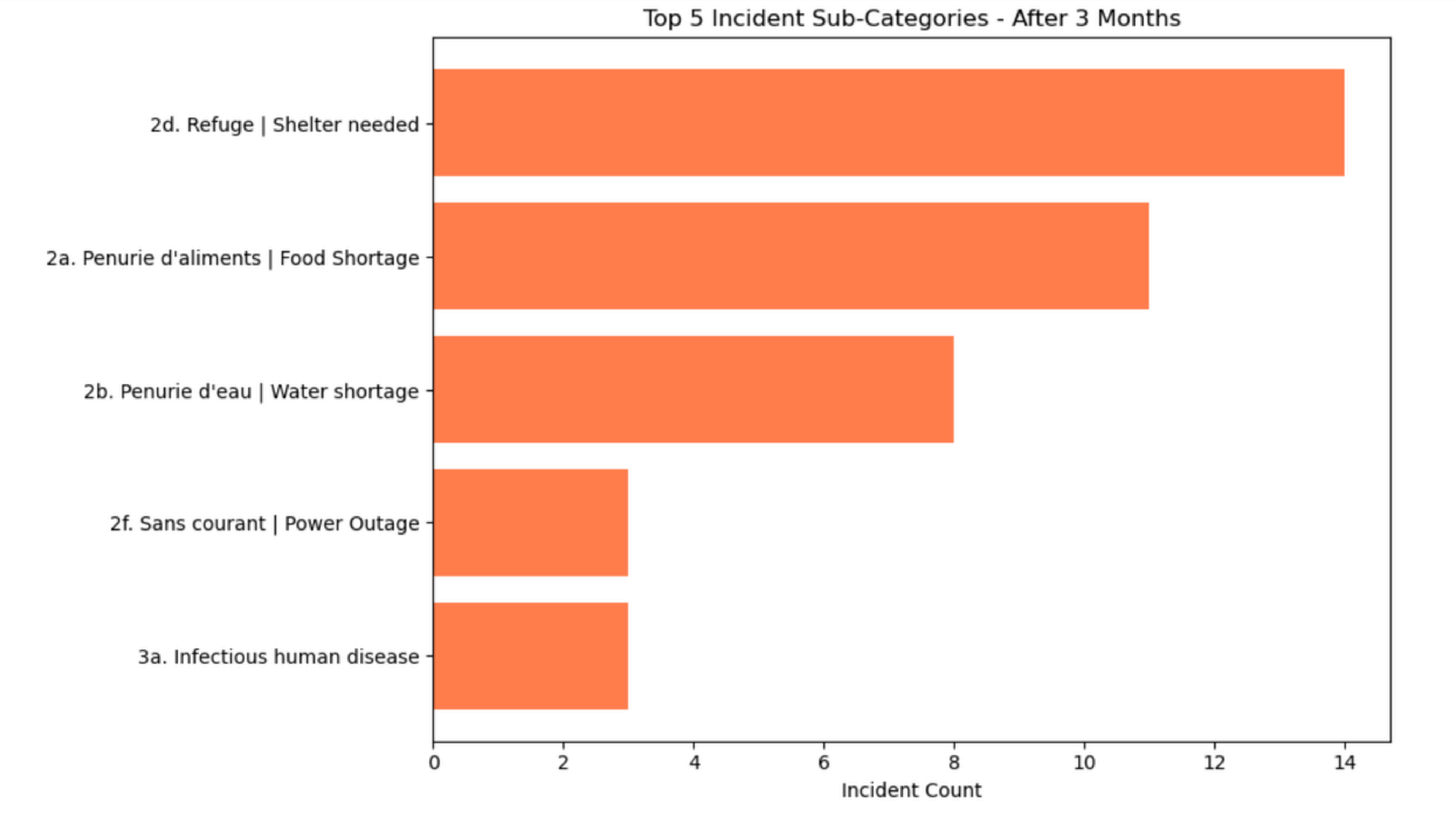


Requests for Basic Needs such as Food and Water were the highest, followed by Means of communication and news, Medical Medical needs.

Prescriptive Analysis

Top 5 Incidents reported after 3 months (Sub-Category)

Request for Refuge was the top reported requests 3 months after the earthquake occurred. Also, Power Outage and request to help in containing Infectious Human Diseases were in the Top 5, indicating there was a change of need for the long term restoration and rehabilitation after the earthquake.



Data Analysis Conclusion

Summary

- The Haiti Dataset offers an interesting analysis of the Earthquake:
- The request/reports were recorded the highest during the first month, however reports continue to come in until June of 2010, indicating the massive effect of this earthquake.
- Reports were approved and responded too, but from a data perspective, it hasn't been verified.
- Proximity affects the number of reports.
- Basic Needs such Food Shortage, Water Shortage and Communication lines were crucial during the first month. Meanwhile, Shelter and combating deseases were issues raised months after the earthquake strike.
- While having a centralized reporting is effective, The Government and other institutions should take note on how to improve the approval and verification of reports.
- The Insights may be used for future policy making especially in Disaster Preparedness and response.

Data Analysis Conclusion

Limitations of Analysis

- Despite having unique kind of data, it seems the dataset is more qualitative than quantitave, limiting the kinds of analysis that can be performed.
- The number of records might be too little to perform specific analysis such as Predictive analysis.
- Since the source of the Data is coming from different channels, there are inconsistencies on how data handled, thus making accurate analysis challenging.

Future Work

- Improve Data Structure for a better Data Analysis
- Add more quantitive data use different analysis methods